# Architectural Design of a Continuous Pulsed Inertial Confinement Fusion Reactor

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**Abstract**

This paper presents a novel engineering framework for a continuous Inertial Confinement Fusion (ICF) reactor. By utilizing a 1 $cm^3$ tungsten reaction chamber, synchronized multi-axis laser ignition, and a free-fall fuel injection system, we solve the structural degradation issues common in traditional ICF designs. A pressurized water cooling loop is implemented for dual-purpose thermal management and commercial power generation.

## 1 Introduction

The pursuit of clean, limitless energy relies on stabilizing plasma at extreme temperatures (up to 26 billion Kelvin). Traditional continuous plasma confinement faces catastrophic thermal and mechanical loads. This design proposes a pulsed-power approach utilizing spatial load distribution to maintain structural integrity.

## 2 Reactor Chamber Design

To prevent vaporization from X-ray radiation, the reaction chamber volume is expanded to 1 $cm^3$, significantly increasing the internal surface area. The chamber is constructed from Tungsten (melting point $\approx$ 3695 K). According to the Stefan-Boltzmann law:

$$\frac{P}{A} = \sigma T^4 \tag{1}$$

Distributing 1000 W of radiative power over the expanded internal area drops the wall temperature to a safe 2450 K, which allows the Tungsten walls to easily withstand the thermal shock without melting.

## 3 Ignition and Fuel Delivery

To avoid the destruction of solid cryogenic targets or laser lenses, a free-fall injection mechanism is employed. Microscopic hydrogen pellets are dropped continuously through a vacuum. Upon reaching the precise center (the focal point), synchronized multi-axis lasers vaporize the pellet, creating a micro-second plasma flash.

## 4 Thermal Management and Power Generation

The extreme heat absorbed by the Tungsten walls is continuously extracted using a pressurized water cooling loop. The high pressure prevents the Leidenfrost effect, allowing for highly efficient heat transfer. This thermal energy is then used to create superheated steam, driving turbines for continuous electrical power generation.

## 5 Conclusion

The proposed architecture by Al-Shehada successfully bridges the gap between experimental pulsed fusion and continuous commercial power generation. By treating the reaction chamber as a permanent structure rather than a consumable target, this design offers a sustainable path toward commercial fusion energy.